

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Easy Integral

Definite Integral with Trigonometric Functions

$$\int_0^{\pi/2} (\cos^{10}(x))(\sin(12x)) dx$$

Solution: We can solve this elegantly by recognizing a pattern in the derivative of a related product of trigonometric functions. Consider the function $f(x) = \cos^{n+1}(x) \cos((n+1)x)$. Let's find its derivative using the product rule and chain rule:

$$\begin{aligned} \frac{d}{dx} [\cos^{n+1}(x) \cos((n+1)x)] &= (n+1) \cos^n(x) (-\sin(x)) \cos((n+1)x) \\ &\quad + \cos^{n+1}(x) (-(n+1) \sin((n+1)x)) \\ &= -(n+1) \cos^n(x) [\sin(x) \cos((n+1)x) + \cos(x) \sin((n+1)x)] \end{aligned}$$

Using the angle addition formula $\sin(A+B) = \sin A \cos B + \cos A \sin B$, the bracketed expression simplifies perfectly to $\sin((n+1)x + x) = \sin((n+2)x)$. Thus, we have the exact differential:

$$\frac{d}{dx} [\cos^{n+1}(x) \cos((n+1)x)] = -(n+1) \cos^n(x) \sin((n+2)x)$$

Rearranging this gives a powerful integration formula:

$$\int \cos^n(x) \sin((n+2)x) dx = -\frac{\cos^{n+1}(x) \cos((n+1)x)}{n+1} + C$$

In our specific problem, we have $n = 10$, which perfectly matches the argument $(n+2)x = 12x$. Applying the formula:

$$\int_0^{\pi/2} \cos^{10}(x) \sin(12x) dx = \left[-\frac{\cos^{11}(x) \cos(11x)}{11} \right]_0^{\pi/2}$$

Evaluating at the upper limit $\frac{\pi}{2}$:

$$-\frac{\cos^{11}(\pi/2) \cos(11\pi/2)}{11} = 0 \quad (\text{since } \cos(\pi/2) = 0)$$

Evaluating at the lower limit 0:

$$-\frac{\cos^{11}(0) \cos(0)}{11} = -\frac{(1)(1)}{11} = -\frac{1}{11}$$

Subtracting the lower limit evaluation from the upper limit evaluation yields our answer:

$$0 - \left(-\frac{1}{11} \right) = \frac{1}{11}$$

1.2 Medium Integral

Algebraic Definite Integral

$$\int_0^1 \frac{1}{\sqrt[3]{4x^2 - 4x + 1} + \sqrt[3]{4x^2 - 1} + \sqrt[3]{4x^2 + 4x + 1}} dx$$

Solution: Let's analyze the expressions inside the cube roots. We can factor them as perfect squares and a difference of squares:

$$4x^2 + 4x + 1 = (2x + 1)^2$$

$$4x^2 - 1 = (2x - 1)(2x + 1)$$

$$4x^2 - 4x + 1 = (2x - 1)^2$$

Let $a = \sqrt[3]{2x + 1}$ and $b = \sqrt[3]{2x - 1}$. The denominator of our integrand cleanly takes the form:

$$a^2 + ab + b^2$$

We can rationalize this denominator by multiplying both the numerator and the denominator by $(a - b)$, exploiting the difference of cubes identity:

$$(a - b)(a^2 + ab + b^2) = a^3 - b^3$$

Substituting our expressions for a and b into $a^3 - b^3$:

$$a^3 - b^3 = (2x + 1) - (2x - 1) = 2$$

Therefore, the integrand simplifies drastically:

$$\frac{1}{a^2 + ab + b^2} = \frac{a - b}{a^3 - b^3} = \frac{\sqrt[3]{2x + 1} - \sqrt[3]{2x - 1}}{2}$$

Now, substitute this back into the integral:

$$\frac{1}{2} \int_0^1 \left((2x + 1)^{1/3} - (2x - 1)^{1/3} \right) dx$$

We can evaluate these integrals using the reverse power rule. Let $u = 2x \pm 1$, then $du = 2 dx$, meaning we multiply by $\frac{1}{2}$ when integrating:

$$\int (2x \pm 1)^{1/3} dx = \frac{1}{2} \cdot \frac{3}{4} (2x \pm 1)^{4/3} = \frac{3}{8} (2x \pm 1)^{4/3}$$

Applying this to our definite integral limits:

$$\frac{1}{2} \left[\frac{3}{8} (2x + 1)^{4/3} - \frac{3}{8} (2x - 1)^{4/3} \right]_0^1 = \frac{3}{16} \left[(2x + 1)^{4/3} - (2x - 1)^{4/3} \right]_0^1$$

Evaluate at the upper limit $x = 1$:

$$(2(1) + 1)^{4/3} - (2(1) - 1)^{4/3} = 3^{4/3} - 1^{4/3} = 3\sqrt[3]{3} - 1$$

Evaluate at the lower limit $x = 0$:

$$(2(0) + 1)^{4/3} - (2(0) - 1)^{4/3} = 1^{4/3} - (-1)^{4/3}$$

Since the real cube root of -1 is -1 , and $(-1)^4 = 1$, we have $(-1)^{4/3} = 1$. Thus, the lower limit evaluation zeroes out:

$$1 - 1 = 0$$

Subtracting the limits, we arrive at our final answer:

$$\frac{3}{16} (3^{4/3} - 1) \quad \text{or} \quad \frac{9\sqrt[3]{3} - 3}{16}$$

1.3 Hard Integral

Evaluating an Integral using Feynman's Technique

Evaluate the Integral:

$$I = \int_0^{\infty} \frac{\ln(1+2x^2)}{(1+x^2)(1+4x^2)} dx$$

Solution: We employ Feynman's technique of integration by introducing a parameter a .

$$I(a) = \int_0^{\infty} \frac{\ln(1+ax^2)}{(1+x^2)(1+4x^2)} dx$$

We want to find $I(2)$. Note that $I(0) = \int_0^{\infty} 0 dx = 0$. Differentiating with respect to a under the integral sign:

$$I'(a) = \int_0^{\infty} \frac{x^2}{(1+ax^2)(1+x^2)(1+4x^2)} dx$$

We perform a partial fraction decomposition on the integrand with respect to $y = x^2$:

$$\frac{y}{(1+ay)(1+y)(1+4y)} = \frac{A}{1+ay} + \frac{B}{1+y} + \frac{C}{1+4y}$$

Using the cover-up method:

$$\begin{aligned} A &= \left. \frac{y}{(1+ay)(1+4y)} \right|_{y=-1/a} = \frac{-1/a}{(1-1/a)(1-4/a)} = \frac{-a}{(a-1)(a-4)} \\ B &= \left. \frac{y}{(1+ay)(1+4y)} \right|_{y=-1} = \frac{-1}{(1-a)(-3)} = \frac{1}{3(1-a)} \\ C &= \left. \frac{y}{(1+ay)(1+y)} \right|_{y=-1/4} = \frac{-1/4}{(1-a/4)(3/4)} = \frac{4}{3(a-4)} \end{aligned}$$

Now, substitute these back into the integral. Recall the standard integral $\int_0^{\infty} \frac{1}{1+ky^2} dy = \frac{\pi}{2\sqrt{k}}$.

$$\begin{aligned} I'(a) &= \int_0^{\infty} \left(\frac{A}{1+ax^2} + \frac{B}{1+x^2} + \frac{C}{1+4x^2} \right) dx \\ &= \frac{\pi}{2} \left(\frac{A}{\sqrt{a}} + B + \frac{C}{\sqrt{4}} \right) \\ &= \frac{\pi}{2} \left(\frac{-\sqrt{a}}{(a-1)(a-4)} + \frac{1}{3(1-a)} + \frac{2}{3(a-4)} \right) \end{aligned}$$

Combining the B and C terms over a common denominator:

$$\frac{1}{3(1-a)} + \frac{2}{3(a-4)} = \frac{1(a-4) + 2(1-a)}{3(1-a)(a-4)} = \frac{-a-2}{3(1-a)(a-4)} = \frac{a+2}{3(a-1)(a-4)}$$

Thus,

$$I'(a) = \frac{\pi}{2} \left(\frac{-\sqrt{a}}{(a-1)(a-4)} + \frac{a+2}{3(a-1)(a-4)} \right) = \frac{\pi}{2} \left(\frac{a-3\sqrt{a}+2}{3(a-1)(a-4)} \right)$$

Notice that the numerator $a - 3\sqrt{a} + 2$ factors as $(\sqrt{a} - 1)(\sqrt{a} - 2)$, and the denominator expands to $3(\sqrt{a} - 1)(\sqrt{a} + 1)(\sqrt{a} - 2)(\sqrt{a} + 2)$. This elegantly cancels out the problematic poles to yield:

$$I'(a) = \frac{\pi}{6(\sqrt{a} + 1)(\sqrt{a} + 2)}$$

Now, we integrate $I'(a)$ from $a = 0$ to $a = 2$ to find $I(2)$:

$$I(2) - I(0) = \frac{\pi}{6} \int_0^2 \frac{1}{(\sqrt{a} + 1)(\sqrt{a} + 2)} da$$

Let $u = \sqrt{a}$, so $da = 2u du$. The limits change from $u = 0$ to $u = \sqrt{2}$.

$$I(2) = \frac{\pi}{3} \int_0^{\sqrt{2}} \frac{u}{(u + 1)(u + 2)} du$$

By partial fractions, $\frac{u}{(u+1)(u+2)} = \frac{-1}{u+1} + \frac{2}{u+2}$.

$$\begin{aligned} I(2) &= \frac{\pi}{3} [2 \ln |u + 2| - \ln |u + 1|]_0^{\sqrt{2}} \\ &= \frac{\pi}{3} \left((2 \ln(\sqrt{2} + 2) - \ln(\sqrt{2} + 1)) - (2 \ln(2) - \ln(1)) \right) \\ &= \frac{\pi}{3} \left(2 \ln \left(\frac{\sqrt{2} + 2}{2} \right) - \ln(\sqrt{2} + 1) \right) \end{aligned}$$

We can simplify this by noting $\frac{\sqrt{2}+2}{2} = \frac{\sqrt{2}}{\sqrt{2}} \left(\frac{1+\sqrt{2}}{\sqrt{2}} \right) = \frac{\sqrt{2}+1}{\sqrt{2}}$. Therefore, $2 \ln \left(\frac{\sqrt{2}+1}{\sqrt{2}} \right) = 2 \ln(\sqrt{2} + 1) - \ln(2)$.

$$\begin{aligned} I(2) &= \frac{\pi}{3} \left(2 \ln(\sqrt{2} + 1) - \ln(2) - \ln(\sqrt{2} + 1) \right) \\ &= \frac{\pi}{3} \left(\ln(\sqrt{2} + 1) - \ln(2) \right) = \frac{\pi}{3} \ln \left(\frac{\sqrt{2} + 1}{2} \right) \end{aligned}$$

Differentials

2.1 Easy Differential

First-Order Linear ODE

Given the differential equation $y' = x + y$, with the initial condition $y(0) = 0$, determine the value of $y(1)$.

Solution: Rearrange the equation into the standard linear form:

$$y' - y = x$$

The integrating factor is $\mu(x) = e^{\int -1 dx} = e^{-x}$. Multiply the entire equation by $\mu(x)$:

$$e^{-x}y' - e^{-x}y = xe^{-x} \implies \frac{d}{dx}(ye^{-x}) = xe^{-x}$$

Integrate both sides with respect to x . The right side requires integration by parts:

$$ye^{-x} = \int xe^{-x} dx = -xe^{-x} - e^{-x} + C$$

Multiplying through by e^x , the general solution is:

$$y(x) = Ce^x - x - 1$$

Apply the initial condition $y(0) = 0$:

$$0 = Ce^0 - 0 - 1 \implies C = 1$$

The particular solution is $y(x) = e^x - x - 1$. Evaluating at $x = 1$:

$$y(1) = e^1 - 1 - 1 = e - 2$$

2.2 Medium Differential

Bernoulli Equation

A micro-drone descending through a storm cloud has altitude $h(t)$. The altitude satisfies:

$$\frac{dh}{dt} + 3h(t) = e^{-t}h(t)^2$$

Given $h(0) = 2$, find the altitude when $t = 1$.

Solution: This is a Bernoulli differential equation of the form $h' + P(t)h = Q(t)h^n$ with $n = 2$. We use the substitution $v = h^{1-n} = h^{-1}$. Differentiating with respect to t yields $v' = -h^{-2}h'$. Divide the original ODE by h^2 :

$$h^{-2}\frac{dh}{dt} + 3h^{-1} = e^{-t}$$

Substitute v and v' :

$$-v' + 3v = e^{-t} \implies v' - 3v = -e^{-t}$$

This is now a first-order linear ODE. The integrating factor is $\mu(t) = e^{\int -3 dt} = e^{-3t}$.

$$\frac{d}{dt}(ve^{-3t}) = -e^{-t}e^{-3t} = -e^{-4t}$$

Integrating both sides:

$$ve^{-3t} = \int -e^{-4t} dt = \frac{1}{4}e^{-4t} + C$$

$$v(t) = \frac{1}{4}e^{-t} + Ce^{3t}$$

Since $v = h^{-1}$, we have $h(t) = \frac{1}{\frac{1}{4}e^{-t} + Ce^{3t}} = \frac{4}{e^{-t} + 4Ce^{3t}}$. Apply the initial condition $h(0) = 2$:

$$2 = \frac{4}{1 + 4C} \implies 1 + 4C = 2 \implies 4C = 1$$

The altitude function is:

$$h(t) = \frac{4}{e^{-t} + e^{3t}}$$

At $t = 1$, the altitude is $h(1) = \frac{4}{e^{-1} + e^3}$.

2.3 Hard Differential

First-Order PDE

A function $u(x, y)$ satisfies the partial differential equation:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = u$$

and becomes $u = 3x - 4$ on the curve $y = 1$. Find the value of $u(4, 2)$.

Solution: We solve this using the Method of Characteristics. The characteristic equations are:

$$\frac{dx}{x} = \frac{dy}{y} = \frac{du}{u}$$

Solving the first pair $\frac{dx}{x} = \frac{dy}{y}$:

$$\ln |x| = \ln |y| + C \implies \frac{x}{y} = C_1$$

Solving the second pair $\frac{dy}{y} = \frac{du}{u}$:

$$\ln |y| = \ln |u| + K \implies \frac{u}{y} = C_2$$

The general solution links the two constants via an arbitrary function f :

$$C_2 = f(C_1) \implies \frac{u}{y} = f\left(\frac{x}{y}\right) \implies u(x, y) = yf\left(\frac{x}{y}\right)$$

We apply the boundary condition $u(x, 1) = 3x - 4$:

$$u(x, 1) = 1 \cdot f\left(\frac{x}{1}\right) = f(x)$$

Thus, the function is $f(x) = 3x - 4$. Substituting this back into the general solution:

$$u(x, y) = y \left(3 \left(\frac{x}{y} \right) - 4 \right) = 3x - 4y$$

We evaluate the specific value at $(4, 2)$:

$$u(4, 2) = 3(4) - 4(2) = 12 - 8 = 4$$

Derivatives

3.1 Easy Derivative

Evaluating a Step Function Derivative

Let $f(x) = \lfloor ((\pi x)^5 + \zeta(\frac{x}{2})\Gamma(\frac{1}{2} + \pi x))^{2x + \cos(\frac{4x\pi}{7})} \rfloor$. Find $f'(\pi + e) + f'(\pi - e) + f'(57)$.

Solution: This problem tests conceptual recognition over computational stamina. The function $f(x)$ represents the floor function, $\lfloor g(x) \rfloor$, where $g(x)$ is a highly complex transcendental expression. The floor function is a step function that maps real numbers to the greatest integer less than or equal to the number. Because the output is constant between integer jumps, the derivative of $\lfloor g(x) \rfloor$ is strictly 0 everywhere it is differentiable (i.e., at all non-integer values of $g(x)$). Given the highly irregular evaluation points $x \in \{\pi + e, \pi - e, 57\}$ plugged into combinations of Gamma, Riemann Zeta, and trigonometric functions, $g(x)$ will map to non-integer values. Consequently, the derivative evaluates exactly to zero at each point.

$$0 + 0 + 0 = 0$$

3.2 Medium Derivative

Logarithmic Differentiation

Let $f(x) = (\sin x)^{\cos x} + (\cos x)^{\sin x}$. Find $f'(\frac{\pi}{4})$.

Solution: Let $u(x) = (\sin x)^{\cos x}$ and $v(x) = (\cos x)^{\sin x}$. We find $u'(x)$ using logarithmic differentiation. $\ln(u) = \cos x \ln(\sin x)$. Differentiating implicitly:

$$\frac{u'}{u} = -\sin x \ln(\sin x) + \cos x \left(\frac{\cos x}{\sin x} \right) = \frac{\cos^2 x}{\sin x} - \sin x \ln(\sin x)$$

$$u'(x) = (\sin x)^{\cos x} \left[\frac{\cos^2 x}{\sin x} - \sin x \ln(\sin x) \right]$$

At $x = \frac{\pi}{4}$, $\sin(\frac{\pi}{4}) = \cos(\frac{\pi}{4}) = \frac{1}{\sqrt{2}}$.

$$\begin{aligned} u'\left(\frac{\pi}{4}\right) &= \left(\frac{1}{\sqrt{2}}\right)^{\frac{1}{\sqrt{2}}} \left[\frac{(1/\sqrt{2})^2}{1/\sqrt{2}} - \frac{1}{\sqrt{2}} \ln\left(\frac{1}{\sqrt{2}}\right) \right] \\ &= \left(\frac{1}{\sqrt{2}}\right)^{\frac{1}{\sqrt{2}}} \left[\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \ln(2^{-1/2}) \right] \\ &= \left(\frac{1}{\sqrt{2}}\right)^{\frac{1}{\sqrt{2}}} \left[\frac{1}{\sqrt{2}} + \frac{1}{2\sqrt{2}} \ln(2) \right] \end{aligned}$$

Similarly for $v(x) = (\cos x)^{\sin x}$: $\ln(v) = \sin x \ln(\cos x)$.

$$\frac{v'}{v} = \cos x \ln(\cos x) + \sin x \left(\frac{-\sin x}{\cos x} \right) = \cos x \ln(\cos x) - \frac{\sin^2 x}{\cos x}$$

$$v'(x) = (\cos x)^{\sin x} \left[\cos x \ln(\cos x) - \frac{\sin^2 x}{\cos x} \right]$$

Evaluating at $x = \frac{\pi}{4}$:

$$\begin{aligned} v' \left(\frac{\pi}{4} \right) &= \left(\frac{1}{\sqrt{2}} \right)^{\frac{1}{\sqrt{2}}} \left[\frac{1}{\sqrt{2}} \ln(2^{-1/2}) - \frac{(1/\sqrt{2})^2}{1/\sqrt{2}} \right] \\ &= \left(\frac{1}{\sqrt{2}} \right)^{\frac{1}{\sqrt{2}}} \left[-\frac{1}{2\sqrt{2}} \ln(2) - \frac{1}{\sqrt{2}} \right] \end{aligned}$$

Summing the derivatives to find $f'(\frac{\pi}{4})$:

$$f' \left(\frac{\pi}{4} \right) = u' \left(\frac{\pi}{4} \right) + v' \left(\frac{\pi}{4} \right) = \left(\frac{1}{\sqrt{2}} \right)^{\frac{1}{\sqrt{2}}} \left[\left(\frac{1}{\sqrt{2}} + \frac{\ln(2)}{2\sqrt{2}} \right) + \left(-\frac{\ln(2)}{2\sqrt{2}} - \frac{1}{\sqrt{2}} \right) \right] = 0$$

3.3 Hard Derivative

Evaluating an Indeterminate Limit

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{21(e^{\sin^{-1}(x)} - e^x)}{\ln(1+x) - \sin(x) + 1 - \cos(x)}$$

Solution: Because applying L'Hôpital's rule would be heavily algebraically taxing, expanding via Maclaurin series around $x = 0$ is the superior approach. We extract terms up to x^3 . For the numerator components:

$$\begin{aligned} \sin^{-1}(x) &= x + \frac{x^3}{6} + O(x^5) \\ e^u &= 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + O(u^4) \\ e^{\sin^{-1}(x)} &= 1 + \left(x + \frac{x^3}{6} \right) + \frac{1}{2} \left(x + \frac{x^3}{6} \right)^2 + \frac{1}{6} \left(x + \frac{x^3}{6} \right)^3 + O(x^4) \\ &= 1 + x + \frac{x^3}{6} + \frac{1}{2}x^2 + \frac{1}{6}x^3 + O(x^4) \\ &= 1 + x + \frac{1}{2}x^2 + \frac{1}{3}x^3 + O(x^4) \end{aligned}$$

Comparing this with the series for e^x :

$$e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + O(x^4)$$

The difference in the numerator is:

$$e^{\sin^{-1}(x)} - e^x \approx \left(\frac{1}{3} - \frac{1}{6} \right) x^3 = \frac{1}{6}x^3$$

Thus, the total numerator expands as $21 \left(\frac{1}{6}x^3 \right) = \frac{7}{2}x^3$.

For the denominator components:

$$\begin{aligned} \ln(1+x) &= x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + O(x^4) \\ \sin(x) &= x - \frac{1}{6}x^3 + O(x^5) \\ 1 - \cos(x) &= 1 - \left(1 - \frac{1}{2}x^2 + O(x^4) \right) = \frac{1}{2}x^2 + O(x^4) \end{aligned}$$

Summing the denominator:

$$\left(x - \frac{1}{2}x^2 + \frac{1}{3}x^3\right) - \left(x - \frac{1}{6}x^3\right) + \frac{1}{2}x^2 = \left(\frac{1}{3} + \frac{1}{6}\right)x^3 = \frac{1}{2}x^3$$

Evaluating the limit as the quotient of leading terms:

$$\lim_{x \rightarrow 0} \frac{\frac{7}{2}x^3}{\frac{1}{2}x^3} = 7$$

Engineering

4.1 Magnetic Field

Magnetic Field of a Current Loop

A circular conducting loop of radius $R = 10$ cm carries a steady current of $I = 5$ A. Determine the magnitude of the magnetic field at a point on the axis of the loop at a distance of $x = 10$ cm from its center in microtesla (μT).
Take $\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$.

Solution: By the Biot-Savart Law, the magnetic field along the axis of a circular loop is given by:

$$B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$$

Substitute the given values, converting to SI units ($R = 0.1$ m, $x = 0.1$ m):

$$(R^2 + x^2)^{3/2} = ((0.1)^2 + (0.1)^2)^{3/2} = (0.02)^{3/2} = (2 \times 10^{-2})^{3/2} = 2\sqrt{2} \times 10^{-3} \text{ m}^3$$

Now, calculate the magnetic field B :

$$\begin{aligned} B &= \frac{(4\pi \times 10^{-7} \text{ T}\cdot\text{m/A})(5 \text{ A})(0.1 \text{ m})^2}{2(2\sqrt{2} \times 10^{-3} \text{ m}^3)} \\ &= \frac{20\pi \times 10^{-9}}{4\sqrt{2} \times 10^{-3}} \text{ T} \\ &= \frac{5\pi}{\sqrt{2}} \times 10^{-6} \text{ T} \end{aligned}$$

Converting to microtesla (μT):

$$B = \frac{5\pi}{\sqrt{2}} \mu\text{T} \approx 11.1 \mu\text{T}$$

4.2 Mechanics

Kinematics of Pure Rolling

A uniform solid cylinder of mass $m = 3$ kg and radius $R = 0.1$ m rests on a long plank of mass $M = 6$ kg. The plank is placed on a frictionless horizontal floor. A constant horizontal force $F = 21$ N is applied to the plank. Assuming pure rolling of the cylinder on the plank without slipping, find the acceleration of the plank in m/s^2 .

Solution: Let a_p be the acceleration of the plank, a_c be the linear acceleration of the cylinder's center of mass, and α be the angular acceleration of the cylinder. Let f be the static friction force acting between the plank and cylinder.

For the plank, applying Newton's Second Law:

$$F - f = Ma_p \implies 21 - f = 6a_p \quad (\text{Eq. 1})$$

For the cylinder's linear motion (friction is the only horizontal force):

$$f = ma_c \implies f = 3a_c \quad (\text{Eq. 2})$$

For the cylinder's rotational motion about its center of mass:

$$\tau = I\alpha \implies fR = \left(\frac{1}{2}mR^2\right)\alpha \implies f = \frac{1}{2}m(R\alpha)$$

Substitute $m = 3 \text{ kg}$:

$$f = 1.5(R\alpha) \quad (\text{Eq. 3})$$

Kinematic condition for pure rolling without slipping: the acceleration of the bottom of the cylinder equals the acceleration of the plank.

$$a_p = a_c + R\alpha \quad (\text{Eq. 4})$$

Equating (Eq. 2) and (Eq. 3):

$$3a_c = 1.5(R\alpha) \implies R\alpha = 2a_c$$

Substitute $R\alpha$ into the kinematic condition (Eq. 4):

$$a_p = a_c + 2a_c = 3a_c \implies a_c = \frac{a_p}{3}$$

Substitute a_c back into (Eq. 2) to express friction in terms of a_p :

$$f = 3\left(\frac{a_p}{3}\right) = a_p$$

Finally, substitute this into the plank's equation (Eq. 1):

$$21 - a_p = 6a_p \implies 7a_p = 21 \implies a_p = 3 \text{ m/s}^2$$

4.3 Quantum

Photoelectric Effect

When photons of wavelength $\lambda_1 = 300 \text{ nm}$ hit a metal target, the maximum kinetic energy is K_1 . When photons of $\lambda_2 = 400 \text{ nm}$ hit the same target, the maximum kinetic energy is K_2 . Given $K_1/K_2 = 2$, find the work function ϕ in eV. Take $hc = 1240 \text{ eV} \cdot \text{nm}$.

Solution: Using the photoelectric equation $K = \frac{hc}{\lambda} - \phi$:

$$K_1 = \frac{1240}{300} - \phi = \frac{124}{30} - \phi = \frac{62}{15} - \phi$$

$$K_2 = \frac{1240}{400} - \phi = \frac{124}{40} - \phi = 3.1 - \phi$$

We are given that $K_1 = 2K_2$:

$$\frac{62}{15} - \phi = 2(3.1 - \phi) = 6.2 - 2\phi$$

Rearrange to solve for the work function ϕ :

$$\phi = 6.2 - \frac{62}{15}$$

Convert 6.2 to a fraction: $6.2 = \frac{62}{10} = \frac{31}{5} = \frac{93}{15}$.

$$\phi = \frac{93}{15} - \frac{62}{15} = \frac{31}{15} \text{ eV} \approx 2.07 \text{ eV}$$